



AI Model Upload Guide

A Developer's Manual for Publishing to the Model Zoo



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INTRODUCTION

Welcome to the SPIDHIVE Model Zoo. This platform serves as the central distribution hub linking our web repository with the mobile application. To ensure compatibility across all devices, every upload must contain a valid **.tflite** model file.

UPLOADING A MODEL



SPIDHIVE powered by SpigTech+

Home Crop Data Set **AI Models** Statistics

Welcome, Nadine Anareta

AI Models [Upload New Model](#)

Models per page: 10

Filters

Keyword Search
Scans model descriptions, names, and tags.

Q e.g. YOLO, Cacao, Pest...

RECENT
banana cacao detect grayscaleinsect rice

Sort By
Date (Newest to Oldest)

Target Crop
Q Search and add a crop...

Model Type
☐ Object Detection
☐ Image Classification

[Clear All Filters](#)

Cirtus Scab Model v2
By Nadine Anareta 4h ago
citrus scab detection
Citrus (1) YOLO11N
146 Images 0 Downloads
PERFORMANCE METRICS
mAP@50: 36.28% mAP@50-95: 15.12%
Precision: 48.6% Recall: 42.42%

Cabbage Cutworm Model v2
By Nadine Anareta 4h ago
cabbage cutworm detection
Cabbage (1) YOLO11N
129 Images 0 Downloads
PERFORMANCE METRICS
mAP@50: 87.89% mAP@50-95: 36.52%
Precision: 88.65% Recall: 91.55%

Corn Gray Leaf Spot Model v2
By Nadine Anareta 16h ago
corn grayleafspot detection
Corn (1) YOLO11N
77 Images 0 Downloads
PERFORMANCE METRICS
mAP@50: 45.74% mAP@50-95: 16.29%
Precision: 51.24% Recall: 51.79%

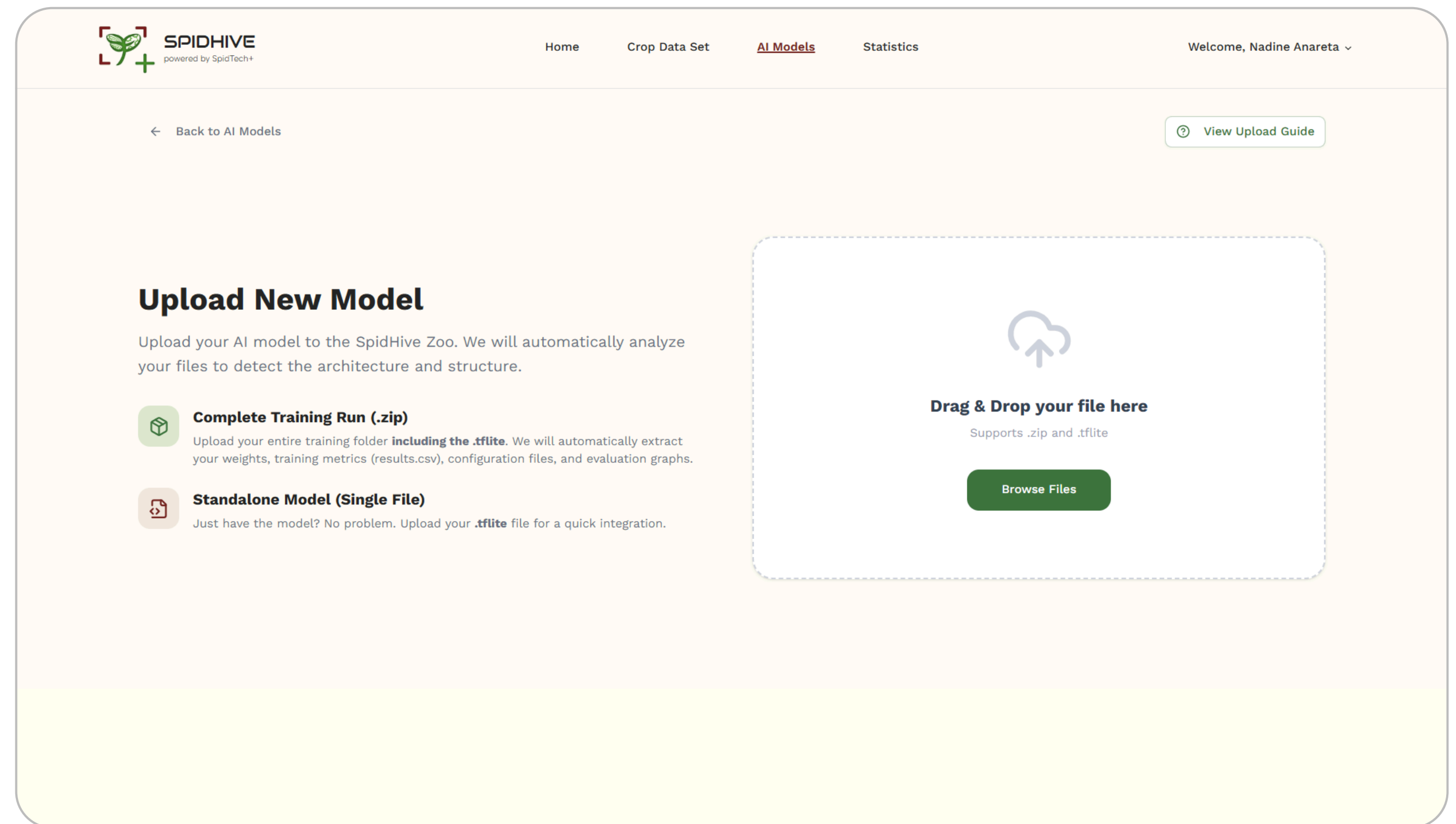
Corn Aphid Detection Version U...
By Nadine Anareta Updated 2d ago
corn aphid detection
Corn (1) YOLO11M
97 Images 0 Downloads
PERFORMANCE METRICS
mAP@50: 20.27% mAP@50-95: 5.22%
Precision: 36.43% Recall: 29.33%

Coffee Green Scale Insect Dete...
By Nadine Anareta Updated 8d ago
coffee detect grayscaleinsect
Coffee (1) YOLO/PyTorch
100 Images 1 Download
PERFORMANCE METRICS
mAP@50: 4.46% mAP@50-95: 1.31%
Precision: 74.94% Recall: 1.86%

Corn Disease - Leaf Spot v2
By Rutherford Belleza Updated 13d ago
corn leafspot
Corn (1) YOLO/PyTorch
77 Images 0 Downloads
PERFORMANCE METRICS
mAP@50: 45.74% mAP@50-95: 16.29%
Precision: 51.24% Recall: 51.79%

Navigate first to the AI Models tab and click on the **“Upload New Model”** button at the top right. This will navigate you to the *Upload* route.

UPLOADING A MODEL



After clicking, you will see a drag-and-drop zone for your model files. You are given two file options: **Zip archive** and **Single File**.

Auto-Extraction



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Home Crop Data Set **AI Models** Statistics Welcome, Nadine Anareta ▾

← Back to AI Models

Publish AI Model

Cancel Input Name

1 Basic Details ⓘ

Model Name *

ex. Cacao Pod Rot YOLOv8

ⓘ Must be unique and clear.

Model Type *

☒ Detection ☐ Classification

Architecture * **Target Crop ***

YOLO11N Select crop...

Base framework. Primary plant subject.

Model Classes

Select specific pests/diseases this model detects.

Select Target Crop first

2 Training Dataset ⓘ

Select the dataset used for training.

Select Target Crop first

7 Documentation *

Write a clear guide for other developers. What does this model do? How was it trained? How can they use it in their code? Use the auto-generate button if you need a head start!

Write Preview Clear All Auto-Generate Upload README.md

B I | | | | <>

Add deployment notes, quirks, or standard Markdown here...

If you trained your model using the YOLO framework, SPIDHIVE can automatically extract your configuration, evaluation graphs, and training metrics to save you time.

How to prepare your file

1. Locate your YOLO training output folder
(typically runs/detect/train/).
2. Ensure the folder contains your **.tflite** weights, **args.yaml**, **results.csv**, and any **.png** evaluation graphs (like confusion matrices).
3. Compress the entire folder into a .zip archive.
4. Drag and drop the .zip into the SPIDHIVE upload zone.



Manual Data Entry



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Home Crop Data Set **AI Models** Statistics Welcome, Nadine Anareta

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Publish AI Model

Cancel Input Name

1 Basic Details

Model Name *
ex. Cacao Pod Rot YOLOv8
Must be unique and clear.

Model Type *
☐ Detection ☒ Classification

Architecture *
Unknown
Base framework.

Target Crop *
Select crop...
Primary plant subject.

Model Classes
Select specific pests/diseases this model detects.
Select Target Crop first

7 Documentation *
Write a clear guide for other developers. What does this model do? How was it trained? How can they use it in their code? Use the auto-generate button if you need a head start!

Write Preview Clear All Auto-Generate Upload README.md

B I | | | | < >

Add deployment notes, quirks, or standard Markdown here...

2 Training Dataset

Select the dataset used for training.

Select Target Crop first

If you are uploading a **standalone** model or using an **architecture other than YOLO** (such as MobileNet or ResNet), you can still publish your model manually.

Steps For Manual Upload



1. Drag and drop your standalone **.tflite** file into the upload zone.
2. In the **"Basic Details"** section, click Upload *args.yaml* to auto-fill your architecture and crop, or simply type them in by hand.
3. In the **"Verified Metrics"** section, click Upload *results.csv* to auto-fill your performance numbers, or manually enter your Final Epoch metrics.
4. In the **"Evaluation Visuals"** section, use the Upload Graphs button to attach any local *.png* or *.jpg* charts.

Best Practices For Documentation



7 Documentation *

Write a clear guide for other developers. What does this model do? How was it trained? How can they use it in their code? Use the auto-generate button if you need a head start!

Write

Preview

Clear All

Auto-Generate

Upload README.md

B I | | | <>

Add deployment notes, quirks, or standard Markdown here...

A great model requires great documentation. In the final step of the publish form, use the Markdown editor to describe:

- **Dataset Lineage:** Which specific dataset was used and what variations it contains.
- **Limitations:** Environments where the model might struggle (e.g., "low light conditions").
- **Parent Models:** If this is a fine-tuned version, link it to its predecessor so users can track the improvement history.